

Diagnosing Separation and Initial-Condition Errors

Answer Key

AP Calculus AB/BC • Error Analysis

Quick Answers

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|------------------|----------|---------|
| 1. $2x^3 - 2x^2$ | 4. 80.34 | 7. 2.61 |
| 2. 13 | 5. 5 | 8. — |
| 3. 14.78 | 6. 2 | |
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Common wrong answers

3. *If they got 4: integrated the right side as if y were a constant instead of separating the variables first*
 4. *If they got 23.09: added the constant after exponentiating instead of carrying it through as a factor*
 5. *If they got 7: took the square root term by term before applying the initial condition*
 6. *If they got 0.67: lost the minus sign from the integral of y to the negative second power*
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1. The right side holds no y , so this integrates directly: $y = \int (6x^2 - 4x) dx$, giving

$y = 2x^3 - 2x^2 + C$

.

The constant selects *which* curve of the family you are on: the slope field is identical for every value of C , and C slides the whole solution curve vertically. Without it there is nothing left for an initial condition to pin down.

2. From problem 1, $y = 2x^3 - 2x^2 + C$. Apply the initial condition *to the general solution*: $y(1) = 2 - 2 + C = 5$, so $C = 5$ and the particular solution is $y = 2x^3 - 2x^2 + 5$. Then $y(2) = 2(8) - 2(4) + 5 = 16 - 8 + 5$, so

$y(2) = 13$

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Watch for: 8, which is this particular solution evaluated with the constant thrown away. Check by differentiating: $\frac{d}{dx}(2x^3 - 2x^2 + 5) = 6x^2 - 4x$ ✓

3. First wrong line: $\int dy = \int xy \, dx$. Wren treated the y on the right as a constant and integrated xy with respect to x . The equation is not yet separated: every y must be moved to the dy side first, because y is a *function of* x and cannot ride along as a coefficient.

Correct solution. Separate, then integrate:

$$\frac{1}{y} dy = x \, dx \Rightarrow \ln |y| = \frac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}.$$

$y(0) = 2$ gives $A = 2$, so $y = 2e^{x^2/2}$ and $y(2) = 2e^2 = 14.7781\dots$, so

$$y(2) \approx 14.78$$

.
Wren's method reports 4. The gap between 4 and 14.78 is the whole cost of not separating: exponential growth was flattened into a parabola.

4. First wrong line: $y = e^{3x} + C$. Exponentiating $\ln |y| = 3x + C$ gives $|y| = e^{3x+C} = e^C e^{3x}$ — the constant rides along as a *factor*, never as a term, because $e^{a+b} = e^a e^b$, not $e^a + b$. Writing $A = e^C$ gives $y = Ae^{3x}$.

Correct solution. $y(0) = 4$ gives $A = 4$, so $y = 4e^{3x}$ and $y(1) = 4e^3 = 80.3421\dots$, so

$$y(1) \approx 80.34$$

.
Dev reports 23.09. Note that his function fails the equation itself: if $y = e^{3x} + 3$ then $y' = 3e^{3x}$ while $3y = 3e^{3x} + 9$, and those differ by 9 for every x — a one-line check that would have caught the error before the arithmetic.

5. Rosa's line breaks two rules at once.

(i) $\sqrt{\frac{y^2}{2}} \neq y$ and $\sqrt{\frac{x^2}{2} + C} \neq x + C$: square roots do not distribute across a sum, so the root cannot be taken term by term.

(ii) The initial condition must be applied to the equation she actually derived, *before* any rearranging that changes what C means. Once C has been dragged through a square root it is no longer the same constant.

Correct solution. From $\frac{y^2}{2} = \frac{x^2}{2} + C$, substitute $x = 0$, $y = 3$: $\frac{9}{2} = C$. Multiply through by 2: $y^2 = x^2 + 9$, so $y = \sqrt{x^2 + 9}$ (positive root, since $y(0) = 3 > 0$). Then $y(4) = \sqrt{16 + 9} = \sqrt{25}$, so

$$y(4) = 5$$

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Rosa's method reports 7 — and 7 is what you get by adding the legs of a right triangle instead of using the hypotenuse, which is the same distributive error seen from a different angle.

6. The missing sign. $\int y^{-2} dy = \frac{y^{-1}}{-1} = -\frac{1}{y}$. Sam dropped the minus, so his first line should read $-\frac{1}{y} = x + C$.

Correct solution. $y(0) = 1$ gives $-1 = 0 + C$, so $C = -1$ and $-\frac{1}{y} = x - 1$, that is $y = \frac{1}{1-x}$. Then $y(0.5) = \frac{1}{0.5}$, so $y(0.5) = 2$.

Behaviour near $x = 1$: the corrected solution blows up — $y \rightarrow \infty$ as $x \rightarrow 1^-$ — so the solution exists only on $(-\infty, 1)$. Sam's version, $y = \frac{1}{x+1}$, is perfectly well behaved there and decays to zero, which hides the vertical asymptote entirely. A sign error changed the qualitative behaviour of the model, not just a digit.

Sam reports 0.67.

7. The right side has no y , so integrate directly. With $u = x^2 + 1$, $du = 2x dx$:

$$y = \int \frac{2x}{x^2 + 1} dx = \ln(x^2 + 1) + C.$$

Apply the initial condition to that general solution: $y(0) = \ln(1) + C = 1$, so $C = 1$ and $y = \ln(x^2 + 1) + 1$. Then $y(2) = \ln 5 + 1 = 2.6094\dots$, so $y(2) \approx 2.61$.

When to apply the condition: immediately after integrating, to the general solution — and never before. Before integrating there is no C in existence to solve for; the initial condition is the instruction that picks one curve out of the family that integration produces.

8. Manual review — expected reasoning. A complete answer does the derivative test and then explains the algebra.

The test. If $y = e^{3x} + 3$ then $\frac{dy}{dx} = 3e^{3x}$, while $3y = 3e^{3x} + 9$. These are equal only if $9 = 0$, so $y = e^{3x} + 3$ is not a solution of $\frac{dy}{dx} = 3y$ at all. More generally $y = e^{3x} + C$ fails for every $C \neq 0$.

The algebra. Solving gives $\ln|y| = 3x + C$, and exponentiating both sides gives $|y| = e^{3x+C} = e^C \cdot e^{3x}$. The constant survives as a *multiplier* $A = e^C$, not as a term, because exponentiation turns sums into products. So the correct family is $y = Ae^{3x}$, and A is exactly as arbitrary as C was — the classmate is right that a constant is arbitrary, and wrong that it can be attached anywhere.

Why it matters: $y = Ae^{3x}$ scales the whole curve, while $y = e^{3x} + C$ shifts it, and only scaling preserves the property that the rate is proportional to the amount.

Accept any answer that (a) differentiates and finds the mismatch of 9, and (b) identifies e^C as a multiplicative constant. Do not accept “you just have to remember to put C first” with no justification.